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TITLE :-

COMPARISON OF SDLC MODELS (WATERFALL, AGILE, SPIRAL, ITERATIVE)

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1. Introduction to SDLC

Software Development Life Cycle (SDLC) is a structured process used to develop high-quality software in a systematic way. It includes various stages such as planning, designing, development, testing, deployment, and maintenance.

SDLC models provide a framework that helps developers and organizations to complete software projects efficiently. Different SDLC models are used based on project requirements, complexity, and risk.

This documentation focuses on comparing four major SDLC models:

- Waterfall Model
- Agile Model
- Spiral Model
- Iterative Model

2. Waterfall Model

Overview

The Waterfall model is a linear and sequential model where each phase must be completed before moving to the next phase.

Characteristics

- Sequential process
- Fixed requirements
- Clear documentation

Advantages

- Easy to understand and manage
- Clear milestones
- Well-structured documentation

Disadvantages

- Not flexible to changes
- Late testing
- Not suitable for complex projects

3. Agile Model

Overview

Agile is an iterative and flexible model that focuses on customer satisfaction and continuous improvement.

Characteristics

- Development in sprints
- Continuous feedback
- Strong team collaboration

Advantages

- Highly flexible
- Faster delivery
- Better customer satisfaction

Disadvantages

- Less documentation
- Requires skilled team
- Difficult to estimate cost

4. Spiral Model

Overview

The Spiral model combines iterative development with risk analysis. It is mainly used for large and complex projects.

Characteristics

- Risk management
- Iterative cycles
- Continuous improvement

Advantages

- Risk reduction
- Early prototypes
- Suitable for high-risk projects

Disadvantages

- Costly
- Time-consuming
- Requires expertise

5. Iterative Model

Overview

The Iterative model develops software in repeated cycles. Each version improves the previous one.

Characteristics

- Step-by-step development
- Early error detection
- Flexible approach

Advantages

- Reduces risk
- Flexible to changes
- Easy to manage

Disadvantages

- Scope may increase
- Needs proper planning
- Incomplete features possible

6. Comparison of Models

Model	Flexibility	Risk Handling	Cost	Best For
Waterfall	Low	Low	Low	Fixed requirements
Agile	High	Medium	Medium	Changing requirements
Spiral	High	High	High	Complex projects
Iterative	Medium	Medium	Medium	Partial requirements

7. Conclusion

In conclusion, each SDLC model has its own advantages and disadvantages. The selection of the model depends on the project requirements.

Waterfall is suitable for simple projects, Agile is best for dynamic projects, Spiral is used for high-risk projects, and Iterative is useful when requirements are not clearly defined.

Choosing the right SDLC model helps in successful project development and better results.